UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC662 mock paper (October 2026). Computed answers show working; theory answers are supported by lecture slides (not lecturer-marked) with a verbatim quote and slide citation.

2. Supported theory marking points quote the lecture decks; any unresolved slot would remain a visible placeholder.

3. Part A = 10 x (2 + 6); Part B = 2 x (5 + 5); total = 100 marks.

4. Computed: 5 slot(s). Supported theory: 19 slots. Placeholders: 0. Questions and answers are for local study use only.

5. Near-copy rejection used all 35 past papers; style widening is recorded per slot.

FOR MARKERS' USE ONLY

This marking scheme consists of 14 printed pages

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CSC662 - MARKING SCHEME (MOCK OCTOBER 2026)

Labels: computed / supported by lecture slides (not lecturer-marked) / placeholder. No official marking scheme exists for the real CSC662 papers; this is generated for the mock.

Entailment labels: supported = the quote directly supports the marking point (or the point passed the stem-overlap gate without an LLM check); partly supported = the quote supports part of the point or is ambiguous; not supported = the quote does not support the point. Only 'not supported' points triggered regeneration.

Part A: 10 x (2 + 6) = 80 marks. Part B: 2 x (5 + 5) = 20 marks. Total: 100 marks. Slots: 5 computed, 19 supported by lecture slides, 0 placeholder.

Methods: computed answers use verified solvers. Theory answers are supported by lecture slides (not lecturer-marked). Each marking point carries a verbatim quote, a deck/slide/chunk citation and a term-overlap score; low overlap is flagged for human review. Sources are restricted to the slot topic's allowed decks, and a slide is not reused across questions where an alternative exists.

PART A - QUESTION 1

A1a - 2 marks - computed - matches lecture slides 5-9 (Caesar shift 3 examples)

Question:

Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

Answer: DEVICERETURNEDANDWIPEDONE

Working:

```
K (10) -7 mod 26 -> D
L (11) -7 mod 26 -> E
C ( 2) -7 mod 26 -> V
P (15) -7 mod 26 -> I
J ( 9) -7 mod 26 -> C
L (11) -7 mod 26 -> E
Y (24) -7 mod 26 -> R
L (11) -7 mod 26 -> E
A ( 0) -7 mod 26 -> T
B ( 1) -7 mod 26 -> U
Y (24) -7 mod 26 -> R
U (20) -7 mod 26 -> N
L (11) -7 mod 26 -> E
K (10) -7 mod 26 -> D
H ( 7) -7 mod 26 -> A
U (20) -7 mod 26 -> N
K (10) -7 mod 26 -> D
D ( 3) -7 mod 26 -> W
P (15) -7 mod 26 -> I
W (22) -7 mod 26 -> P
L (11) -7 mod 26 -> E
K (10) -7 mod 26 -> D
V (21) -7 mod 26 -> O
U (20) -7 mod 26 -> N
L (11) -7 mod 26 -> E
```

Suggested marks: 1 mark for identifying the cipher/shift direction; 1 mark for the complete plaintext.
Confidence: computed - matches lecture slides 5-9 (Caesar shift 3 examples).

A1b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six facts about substitution ciphers as presented in the lecture, covering how plain-text is transformed, what serves as the key, the security principle involved, and how the Vigenere cipher differs from simple substitution.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment

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1	Plain-text letters are swapped for other letters to produce the cipher-text.	"In substitution ciphers, we replace the plain-text letters with other letters"	031 - Classical Cryptography, slide 4	031 - Classical Cryptography::s0 4::1	0.71	supported (ste gate)
2	The rule governing the replacement acts as the cipher key.	"The substitution rule is the key"	031 - Classical Cryptography, slide 4	031 - Classical Cryptography::s0 4::1	0.20 LOW	supported
3	Substitution embodies Shannon's confusion principle and this concept carries over into modern ciphers.	"Corresponds to Shannon's principle of confusion"	031 - Classical Cryptography, slide 4	031 - Classical Cryptography::s0 4::1	0.27 LOW	partly supporte
4	The Vigenere scheme is a polyalphabetic cipher that extends Caesar's shift cipher.	"Vigenere's polyalphabetic cipher (19th century) generalizes Caesar's shift cipher"	031 - Classical Cryptography, slide 14	031 - Classical Cryptography::s1 4::1	0.71	supported (ste gate)
5	A keyword is used to choose which encrypting rows are applied.	"Use keyword to select encrypting rows"	031 - Classical Cryptography, slide 14	031 - Classical Cryptography::s1 4::1	0.60	supported (ste gate)
6	The Vigenere cipher resists straightforward frequency analysis.	"The Vigenere cipher is not amenable to simple frequency analysis"	031 - Classical Cryptography, slide 14	031 - Classical Cryptography::s1 4::1	0.67	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 031 - Classical Cryptography, slide 14; 031 - Classical Cryptography, slide 4

PART A - QUESTION 2

A2a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how a rail fence transposition cipher encrypts and decrypts a message.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Encryption writes the plaintext in a zigzag across rows and reads letters row by row to form the ciphertext.	"The plain-text is written in a zigzag pattern in rows and form the cipher-text by reading off the letters from the rows."	031 - Classical Cryptography, slide 25	031 - Classical Cryptography::s2 5::1	0.50	supported (ste gate)
2	Decryption rebuilds the empty rail fence table from the known key, fills rows with ciphertext, then reads in zigzag to recover plaintext.	"To decrypt a rail fence cipher, we have to recreate the 'empty' rail fence table (as in (A)) using the known key."	031 - Classical Cryptography, slide 26	031 - Classical Cryptography::s2 6::1	0.40 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 031 - Classical Cryptography, slide 25; 031 - Classical Cryptography, slide 26

A2b - 6 marks - computed - matches lecture slide 15 (standard tableau)

Question:

Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

Answer: SECURELOGINNOW

The 26 x 26 Vigenere tableau is printed in the question paper.

Working:

```
Standard 26 x 26 tableau, A=0; key repeated over the ciphertext.
Ciphertext: UMRBVVNWVPREQE
Key:        CIPHERCIPHERCI
Letter-by-letter: C - K mod 26 = P
```

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```
  U (20) - C ( 2) = 18 -> S
  M (12) - I ( 8) =  4 -> E
  R (17) - P (15) =  2 -> C
  B ( 1) - H ( 7) = 20 -> U
  V (21) - E ( 4) = 17 -> R
  V (21) - R (17) =  4 -> E
  N (13) - C ( 2) = 11 -> L
  W (22) - I ( 8) = 14 -> O
  V (21) - P (15) =  6 -> G
  P (15) - H ( 7) =  8 -> I
  R (17) - E ( 4) = 13 -> N
  E ( 4) - R (17) = 13 -> N
  Q (16) - C ( 2) = 14 -> O
  E ( 4) - I ( 8) = 22 -> W
Plaintext:  SECURELOGINNOW
```

Suggested marks: 1 mark for the key/table setup; 4 marks for the letter-by-letter working; 1 mark for the complete answer.

Confidence: computed - matches lecture slide 15 (standard tableau).

PART A - QUESTION 3

Status: computed placeholder - no question generated for this type.

A3b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Compare the columnar transposition cipher with the double transposition variant, covering how each encrypts, what serves as the key, and how the two differ in output.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Encryption under columnar transposition is performed by writing the plain-text into matrix rows and then reading the cipher-text out of columns.	"To encrypt - put plain-text into rows of matrix then read cipher-text out of columns"	031 - Classical Cryptography, slide 20	031 - Classical Cryptography::s20::1	0.69	supported (ste gate)
2	The lecture illustrates columnar transposition with a 3 x 4 matrix applied to the plain-text ATTACK, yielding the cipher-text ACXTKXTXXAXX.	"suppose matrix is 3 x 4 and the plain-text is ATTACK"	031 - Classical Cryptography, slide 20	031 - Classical Cryptography::s20::1	0.36 LOW	partly supporte
3	In the double transposition example the key-space is stated as 4 and the keyword used is MATH.	"The key-space is 4 and the keyword is MATH"	031 - Classical Cryptography, slide 24	031 - Classical Cryptography::s24::1	0.50	supported (ste gate)
4	Double transposition adds a second stage in which the keyword is sorted to AHMT before the final cipher-text is produced.	"Then we perform 2nd transposition by sorting the keyword to AHMT."	031 - Classical Cryptography, slide 24	031 - Classical Cryptography::s24::1	0.31 LOW	supported
5	The final cipher-text obtained from the double transposition example is TKXAXXACXTXX.	"Finally, the cipher-text is TKXAXXACXTXX."	031 - Classical Cryptography, slide 24	031 - Classical Cryptography::s24::1	0.50	supported (ste gate)
6	The lecture frames the double transposition as a keyword-based columnar transposition cipher, distinguishing it from the plain columnar case.	"Keyword Columnar Transposition Ciphers"	031 - Classical Cryptography, slide 24	031 - Classical Cryptography::s24::1	0.40 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited

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lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 031 - Classical Cryptography, slide 20; 031 - Classical Cryptography, slide 24

PART A - QUESTION 4

A4a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) formal security models and the specific security concern each addresses.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The Bell-La Padula (BLP) model is a formal security model that deals with access control only.	"Bell-La Padula (BLP) model: access control only"	024 - Formal Security Model, slide 5	024 - Formal Security Model::s05::1	0.70	supported (ste gate)
2	The Chinese Wall model is a formal security model designed to identify conflicts of interest.	"Chinese Wall model: identification of conflicts of interest"	024 - Formal Security Model, slide 5	024 - Formal Security Model::s05::1	0.62	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 024 - Formal Security Model, slide 5

Status: computed placeholder - no question generated for this type.

PART A - QUESTION 5

A5a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe two ways substitution is used in classical encryption.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Encryption historically and today involves substitution as one of its operations.	"Historically, and also today, encryption involves"	031 - Classical Cryptography, slide 2	031 - Classical Cryptography::s02::1	0.67	supported (ste gate)
2	An early documented substitution method has women encoding love messages by replacing letters with pre-paired letters.	"Kama-Sutra suggests that women learn to encrypt their love messages by substituting pre-paired letters (4th Century AD)"	031 - Classical Cryptography, slide 2	031 - Classical Cryptography::s02::1	0.54	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 031 - Classical Cryptography, slide 2

A5b - 6 marks - computed - 3 x 4 fill-rows/read-columns convention verified against lecture slides 20-21 and 24

Question:

Decrypt the following ciphertext using columnar transposition with a 4 x 6 matrix (4 rows, 6 columns). Put the ciphertext into the columns, read the plaintext out of the rows, and ignore the filler X at the end. The key is the keyword secret.
FOIENCXLRPYXOMTSEEEUCLRX

Answer: ENFORCECOMPLEXITYRULES

Working:

```
  Keyword: SECRET; read order: col 3 (C), col 2 (E), col 5 (E), col 4 (R), col 1 (S), col
6 (T)
  Grid: 6 columns x 4 rows (24 cells; ciphertext 24 chars; transport pad 0)
  Ciphertext split in read order:
    col 3 (C): FOIE
    col 2 (E): NCXL
    col 5 (E): RPYX
    col 4 (R): OMTS
    col 1 (S): EEEU
    col 6 (T): CLRX
  Reconstructed rows:
    row 1: ENFORC
    row 2: ECOMPL
    row 3: EXITYR
    row 4: ULESXX
  Row-wise: ENFORCECOMPLEXITYRULESXX
  Remove 2 trailing pad 'X': ENFORCECOMPLEXITYRULES
```

Suggested marks: 1 mark for the key column order; 2 marks for reconstructing the columns; 2 marks for the row-wise plaintext; 1 mark for stripping the padding.

Confidence: computed - 3 x 4 fill-rows/read-columns convention verified against lecture slides 20-21 and 24.

PART A - QUESTION 6

A6a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) key sources of software insecurity.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Attackers can tamper with system resources that are critical to program execution, such as the instruction pointer or function call addresses.	"Software insecurity stems from attacker's ability to modify system resources criticial for program execution, e.g. instruction pointer, function call addresses, interrupt addresses, etc."	051 - Software Security, slide 30	051 - Software Security::s30::1	0.71	supported (ste gate)
2	A major cause of software insecurity is the failure to properly validate input supplied by users.	"One of the key sources for software insecurity is failed validation of user input."	051 - Software Security, slide 30	051 - Software Security::s30::1	0.50	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.

Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).

Citations: 051 - Software Security, slide 30

A6b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six (6) aspects of Windows security mechanisms, covering their purpose, the entities they manage, and the risks arising from their complexity.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Windows provides elaborate built-in facilities for managing user accounts and objects and for controlling access.	"Windows systems contain complex security mechanisms for user account and object management as well as access control."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.64	supported (ste gate)
2	Discretionary access control in Windows permits detailed, fine-grained control over access to diverse entities both within the	"DACS enable fine-grained access control of heterogeneous entities in Windows itself as well as applications."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.43 LOW	supported

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	OS and in applications.					
3	The very complexity of these mechanisms creates danger, since incorrect configuration or coding defects can cause serious security breaches.	"The complexity of Windows security mechanisms is also its enemy: misconfiguration as well as implementation bugs may lead to severe security incidents."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.20 LOW	supported
4	Processes and threads act as the active subjects on which Windows enforces its security.	"Windows subjects are processes and threads."	041 - Windows Security, slide 11	041 - Windows Security::s11::1	0.50	supported (ste gate)
5	A subject's security credentials are held in a token that maps the principal to the subject and can carry extra security attributes.	"Security credentials for a subject are stored in a token."	041 - Windows Security, slide 11	041 - Windows Security::s11::1	0.36 LOW	partly supporte
6	Windows 2000 introduced Active Directory, an LDAP-like directory service that organizes users, groups, credentials, resources and security policies.	"Active directory introduced in Windows 2000 is an LDAP-like directory service for organization of system resources:"	041 - Windows Security, slide 6	041 - Windows Security::s06::1	0.67	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 041 - Windows Security, slide 11; 041 - Windows Security, slide 21; 041 - Windows Security, slide 6

PART A - QUESTION 7

A7a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain two characteristics of Trojan horse malware that distinguish it from other malicious programs.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Trojans disguise themselves as useful or benign software in order to get onto the victim's system.	"They usually masquerade as helpful or harmless programs"	051 - Software Security, slide 12	051 - Software Security::s12::1	0.00 LOW	supported
2	Once running, a Trojan steals private data, deletes files, records keystrokes, or opens a back door for further malware.	"begin to steal private data, delete files, log keystrokes or provide an opening for further malware to get installed"	051 - Software Security, slide 12	051 - Software Security::s12::1	0.60	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 12

A7b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six techniques by which malware infects systems or propagates itself, including how it hides its activity and how it spreads through user interaction.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Boot sector infection works by replacing the original master boot	"Overwrite the MBR with an infected one"	051 - Software Security, slide 8	051 - Software Security::s08::1	0.15 LOW	supported

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	record with an infected version so the system bootstraps using the malicious MBR.					
2	COM executable infection appends the virus body to a program and replaces the program entry point with a jump to a virus body.	"Replace a program entry point with a jump to a virus body"	051 - Software Security, slide 9	051 - Software Security::s09::1	0.64	supported (ste gate)
3	EXE executable infection overwrites a program header to switch the entry point to a virus.	"Over write a program header to switch the entry point to a virus"	051 - Software Security, slide 10	051 - Software Security::s10::1	0.60	supported (ste gate)
4	Rootkits conceal malicious processes and software in the infected system to keep privileged remote access for the author.	"Rootkits are a type of malware that mask malicious processes and software in the infected systems to ensure privileged remote access."	051 - Software Security, slide 13	051 - Software Security::s13::1	0.75	supported (ste gate)
5	Spyware propagates harm by spying on users, stealing logins and passwords and logging keystrokes.	"spyware are programs that spy on the users. They steal user logins and passwords, document sites visited, log keystrokes and even redirect browsers to unwanted sites."	051 - Software Security, slide 13	051 - Software Security::s13::1	0.80	supported (ste gate)
6	Cross-site scripting spreads through user interaction when a victim inadvertently clicks an emailed link or an infected image.	"Inadvertently click on an emailed link"	052 - Web Security, slide 13	052 - Web Security::s13::1	0.27 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 10; 051 - Software Security, slide 13; 051 - Software Security, slide 8; 051 - Software Security, slide 9; 052 - Web Security, slide 13

PART A - QUESTION 8

A8a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List TWO (2) stages involved in the biometric authentication process.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The identification stage, which involves acquiring the biometric and creating and storing master characteristics.	"Identification Stage Acquisition Creation of Master characteristics Storage of Master characteristics"	072 - Biometrics and Authentication, slide 5	072 - Biometrics and Authentication::s05::1	0.38 LOW	supported
2	The authentication stage, which involves further acquisition, comparison and a decision.	"Authentication Stage Acquisition(s) Comparison Decision"	072 - Biometrics and Authentication, slide 5	072 - Biometrics and Authentication::s05::1	0.71	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 072 - Biometrics and Authentication, slide 5

A8b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain six (6) points about biometric mechanisms and technologies, covering how a biometric is defined,

the categories of biometric applications, and the strengths and weaknesses of specific biometric technologies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A biometric is defined as a physiological or behavioural characteristic of a human being that can distinguish one person from another.	"A biometric is a physiological or behavioural characteristic of a human being that can distinguish one person from another"	072 - Biometrics and Authentication, slide 2	072 - Biometrics and Authentication::s02::1	0.89	supported (ste gate)
2	Biometric applications are grouped into two sectors, namely psychological and behavioural.	"Biometric applications available today are categorized into 2 sectors"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.57	supported (ste gate)
3	The psychological category includes iris, fingerprints, hand, retinal and face recognition.	"Psychological: Iris, Fingerprints, Hand, Retinal and Face recognition"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.78	supported (ste gate)
4	The behavioural category includes voice, typing pattern and signature.	"Behavioural: Voice, Typing pattern, Signature"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.67	supported (ste gate)
5	Iris scanning reads the unique pattern of the iris, needs no touch, and overcomes the issues of retinal scanners.	"Scans unique pattern of iris"	072 - Biometrics and Authentication, slide 10	072 - Biometrics and Authentication::s10::1	0.36 LOW	partly supporte
6	Fingerprint technology suffers from disadvantages such as racial issues, dirt, grime and wounds, finger placement, an oversized database, and spoofing that makes liveness important.	"Can be spoofed - liveness important!"	072 - Biometrics and Authentication, slide 7	072 - Biometrics and Authentication::s07::1	0.17 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 072 - Biometrics and Authentication, slide 10; 072 - Biometrics and Authentication, slide 2; 072 - Biometrics and Authentication, slide 4; 072 - Biometrics and Authentication, slide 7

PART A - QUESTION 9

A9a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the access control list (ACL) and access control matrix (ACM) structures.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	An ACL describes access rights from the object's perspective, listing which subjects hold which rights on that object.	"ACL is an object-centered description of access rights:"	023 - Access Control and Authorization, slide 10	023 - Access Control and Authorization::s10::1	0.57	supported (ste gate)
2	The access control matrix is a basic control structure whose strengths are clarity of definition and ease of verification.	"ACM is a basic control structure."	023 - Access Control and Authorization, slide 8	023 - Access Control and Authorization::s08::1	0.27 LOW	partly supporte

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Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 023 - Access Control and Authorization, slide 10; 023 - Access Control and Authorization, slide 8

A9b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Refer to your answer in (a). Explain the goals of access control and the role of the reference monitor, then describe two formal security models and the specific concern each addresses, and explain the ACL and ACM structures.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Access control has four goals: granting access, limiting access, preventing access and revoking access.	"Goals of access control: Granting access Limiting access Preventing access Revoking access"	023 - Access Control and Authorization, slide 4	023 - Access Control and Authorization::s04::1	0.88	supported (stegate)
2	The reference monitor is an abstract mechanism that controls access requests.	"Reference monitor is an abstract notion of a mechanism for controlling access requests."	023 - Access Control and Authorization, slide 6	023 - Access Control and Authorization::s06::1	1.00	supported (stegate)
3	Reference monitors are instruments for realizing access control policies and can be deployed at all levels of the system hierarchy.	"Reference monitors are instruments for realization of access control policies."	023 - Access Control and Authorization, slide 18	023 - Access Control and Authorization::s18::1	0.54	supported (stegate)
4	The Chinese Wall model addresses conflicts of interest for a consultant or contractor.	"Brewer and Nash's Chinese Wall Policy is designed to address a very specific concern: conflicts of interest by a consultant or contractor."	024 - Formal Security Model, slide 27	024 - Formal Security Model::s27::1	0.88	supported (stegate)
5	The Chinese Wall is an access control confidentiality policy, not strictly an integrity policy.	"Strictly speaking, this is not an integrity policy, but an access control confidentiality policy."	024 - Formal Security Model, slide 21	024 - Formal Security Model::s21::1	0.78	supported (stegate)
6	The access control matrix and access control lists are structures for implementing access policies.	"Access control structures are mechanisms for implementing access policies: access control matrix (ACM) Capabilities (C-List) access control lists (ACL)"	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	1.00	supported (stegate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 023 - Access Control and Authorization, slide 18; 023 - Access Control and Authorization, slide 4; 023 - Access Control and Authorization, slide 6; 023 - Access Control and Authorization, slide 7; 024 - Formal Security Model, slide 21; 024 - Formal Security Model, slide 27

PART A - QUESTION 10

A10a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how a chain of trust is established in trusted computing.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Each stage of the chain measures and validates the following stage.	"Each stage measures and validates the next one"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.60	supported (stegate)
2	The resulting measurements are recorded in Platform Configuration Registers on the TPM.	"Measurements go into Platform Configuration Registers (PCRs) on the TPM"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.71	supported (stegate)

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Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 073 - Trusted Computing, slide 8

A10b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss the NGSCB architecture by describing the roles of its Nexus and Nexus Computing Agent components and how they relate to the TPM. Provide six distinct points.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Nexus is described as the security kernel that is authenticated during boot.	"Security kernel, authenticated on boot"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.67	supported (ste gate)
2	Nexus authenticates trusted programs running on the platform.	"Authenticates trusted programs"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.50	supported (ste gate)
3	Nexus provides the application interface to the TPM.	"Application interface to TPM"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.60	supported (ste gate)
4	Nexus does not place trust in the operating system.	"Does not trust OS"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.43 LOW	supported
5	The Nexus Computing Agent is a trusted software component that runs in trusted mode and communicates with Nexus.	"Runs in trusted mode that communicates with Nexus"	073 - Trusted Computing, slide 15	073 - Trusted Computing::s15::1	0.62	supported (ste gate)
6	NCA specifications are open-source, allowing developers to build their own agents for the trusted platform.	"Developers can make their own agents to run on the trusted platform"	073 - Trusted Computing, slide 15	073 - Trusted Computing::s15::1	0.50	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 073 - Trusted Computing, slide 14; 073 - Trusted Computing, slide 15

PART B - QUESTION 1

B1a - 5 marks - supported by lecture slides (not lecturer-marked)

Scenario:

A computing professional at a company handles personal information about customers. The professional must decide how to act, guided by morals, ethics and laws, and by the ACM code of ethics and professional conduct.

Question:

Discuss how morals, ethics and laws differ, and explain how the ACM code guides the professional's conduct in the scenario.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	State that morals are traditions of belief about right and wrong conduct, forming the bedrock of social behaviour.	"Morals are traditions of belief about right and wrong conduct and "doing what is morally right" is the bedrock of our social behavior"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	0.82	supported (ste gate)
2	Explain that ethics is a suite of guiding beliefs, standards or ideals that pervades an individual or a group or community.	"Ethics is a suite of guiding beliefs, standards, or ideals that pervades an individual or a group or community of people"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	1.00	supported (ste gate)

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3	Describe laws as formal rules of conduct imposed by a sovereign authority such as a government on its subjects or citizens.	"Laws are formal rules of conduct that a sovereign authority, such as a government, imposes on its subjects or citizens"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	1.00	supported (ste gate)
4	Note that the ACM code requires the professional to respect privacy.	"Respect privacy."	071 - Ethics and Computer Security, slide 11	071 - Ethics and Computer Security::s11::1	0.29 LOW	supported
5	Note that the ACM code requires the professional to honor confidentiality.	"Honor confidentiality."	071 - Ethics and Computer Security, slide 11	071 - Ethics and Computer Security::s11::1	0.29 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 11; 071 - Ethics and Computer Security, slide 5

B1b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain five ways the ACM code of ethics and professional conduct should shape the professional's behaviour when handling customer personal information in the scenario.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The professional should aim for high quality in the processes and products of the work done with the customer data.	"Strive to achieve high quality in both the processes and products of professional work."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.55	supported (ste gate)
2	The professional should uphold strong standards of competence, conduct and ethical practice while handling the information.	"Maintain high standards of professional competence, conduct, and ethical practice."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.50	supported (ste gate)
3	The professional should be aware of and respect the rules that apply to the professional work involving the data.	"Know and respect existing rules pertaining to professional work."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.44 LOW	supported
4	The professional should only access computing and communication resources when authorised or when the public good requires it.	"Access computing and communication resources only when authorized or when compelled by the public good."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.67	supported (ste gate)
5	The professional should treat the code as a documented set of acceptable behaviours defined by the professional group, identifying which actions are appropriate or inappropriate.	"Documented set of acceptable behaviors for professional or social groups."	071 - Ethics and Computer Security, slide 6	071 - Ethics and Computer Security::s06::1	0.39 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 12; 071 - Ethics and Computer Security, slide 6

PART B - QUESTION 2

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B2a - 5 marks - supported by lecture slides (not lecturer-marked)

Scenario:

A security analyst is reviewing an incident in which an attacker injected JavaScript through a user request into a web application, and separately a program spread itself across machines on the network. The analyst must classify each event by the class of threat it represents and by the security objective that is affected.

Question:

Explain how the classes of threats map to impacts on confidentiality, integrity and availability, giving one distinct impact for each security objective.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Disclosure is unauthorized access to information, which harms confidentiality.	"Disclosure - unauthorized access to information"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.67	supported (ste gate)
2	Deception involves acceptance of false data, which harms integrity.	"Deception - acceptance of false data"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.57	supported (ste gate)
3	Disruption interrupts or prevents correct operation, which harms integrity.	"Disruption - interruption or prevention of correct operation"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.71	supported (ste gate)
4	Usurpation is unauthorized control of part of a system, and availability services counter it.	"Usurpation - unauthorized control of some part of a system"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.62	supported (ste gate)
5	Protection mechanisms are classified according to five security objectives including confidentiality, integrity and availability.	"Protection mechanisms are classified according to 5 security objectives - Confedentiality, Integrity, Availability, Authentication and Accountability"	011 - Introduction to Computer Security, slide 28	011 - Introduction to Computer Security::s28::1	0.70	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 011 - Introduction to Computer Security, slide 10; 011 - Introduction to Computer Security, slide 28

B2b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Referring to the scenario, state five facts about the Samy worm that show how an XSS payload can propagate itself and what impact it had.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The worm injected JavaScript into a user profile using document.body.innerHTML.	"Inject Javascript into a user profile using document.body.innerHTML"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.78	supported (ste gate)
2	It obtained the list of the user's heroes by issuing a GET request on a user profile.	"Get the list of user's heros by using a GET on a user profile"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.43 LOW	supported
3	It added Samy as a friend by performing XML-HTTP on the addFriends page.	"Add Samy as a field by performing XML-HTTP on the addFriends page"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.88	supported (ste gate)
4	Any visitor to an infected user's page had their own profile and friends list infected, so the worm spread through page visits.	"Any user who visits an infected user's page will gets profile infected with Javascript and his friends list infected with Samy"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.46 LOW	supported

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5	The outbreak eventually infected about 1,000,000 MySpace profiles.	"Eventually about 1,000,000 MySpace profiles were infected."	052 - Web Security, slide 15	052 - Web Security::s15::1	0.71	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 052 - Web Security, slide 15

End of marking scheme.

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